

1D Arrays (Linear Arrays)

1. Sum of Array Elements

- Write a program that takes an integer array as input and returns the sum of all its elements.
- **Example:** For an array `[1, 2, 3, 4, 5]`, the output should be `15`.

2. Find Maximum Element in Array

- Write a function that finds and returns the maximum element in an integer array.
- **Example:** For an array `[3, 5, 7, 2, 8]`, the output should be `8`.

3. Reverse an Array

- Write a method that reverses the elements of a given integer array.
- **Example:** For an array `[1, 2, 3, 4, 5]`, the output should be `[5, 4, 3, 2, 1]`.

4. Count Occurrences of an Element

- Write a program that takes an integer array and an integer value as input and returns the count of occurrences of that value in the array.
- **Example:** For an array `[1, 2, 3, 2, 4, 2]` and value `2`, the output should be `3`.

5. Check if Array is Sorted

- Write a function that checks if a given array is sorted in ascending order.
- **Example:** For an array `[1, 2, 3, 4, 5]`, the output should be `true`. For `[1, 3, 2]`, the output should be `false`.

6. Remove Duplicates from Array

- Write a program to remove duplicate elements from an integer array and print the unique elements.
- **Example:** For an array `[1, 2, 2, 3, 4, 4, 5]`, the output should be `[1, 2, 3, 4, 5]`.

7. Shift Array Elements to the Right

- Write a program that takes an array and shifts each element to the right by one position, with the last element moving to the first position.
- **Example:** For `[1, 2, 3, 4, 5]`, the output should be `[5, 1, 2, 3, 4]`.

8. Find Pairs with Given Sum

- Write a function to find all pairs in an array that sum up to a given number.
- **Example:** For `[1, 2, 3, 4, 5]` and sum `5`, the output should be pairs `(1, 4)` and `(2, 3)`.

9. Find the Second Largest Element

- Write a program that finds the second largest element in an integer array.
- **Example:** For `[3, 5, 7, 2, 8]`, the output should be `7`.

10. Move Zeros to the End

- Write a function that moves all zeros in an array to the end while maintaining the order of non-zero elements

- **Example:** For `[0, 1, 0, 3, 12]`, the output should be `[1, 3, 12, 0, 0]`.
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2D Arrays (Matrix)

1. Sum of Elements in 2D Array

- Write a program to calculate the sum of all elements in a 2D integer array.
- **Example:** For a 2D array `{{1, 2}, {3, 4}}`, the output should be `10`.

2. Transpose of a Matrix

- Write a function that takes a 2D array (matrix) as input and returns its transpose.
- **Example:** For `{{1, 2}, {3, 4}}`, the output should be `{{1, 3}, {2, 4}}`.

3. Row-wise Sum of 2D Array

- Write a program that calculates the sum of elements in each row of a 2D array and prints it.
- **Example:** For `{{1, 2, 3}, {4, 5, 6}}`, the output should be `6` for the first row and `15` for the second.

4. Check if Matrix is Symmetric

- Write a function that checks if a given square matrix is symmetric (i.e., the matrix is equal to its transpose).
- **Example:** For `{{1, 2, 3}, {2, 1, 4}, {3, 4, 1}}`, the output should be `true`.

5. Diagonal Sum of Square Matrix

- Write a program to find the sum of the diagonal elements of a square matrix.
- **Example:** For `{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}`, the output should be `15` (sum of 1, 5, and 9).

6. Rotate Matrix by 90 Degrees

- Write a function to rotate a given 2D matrix by 90 degrees clockwise.
- **Example:** For `{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}`, the output should be `{{7, 4, 1}, {8, 5, 2}, {9, 6, 3}}`.

7. Sum of Each Column in a 2D Array

- Write a program to calculate the sum of elements in each column of a 2D array and print it.
- **Example:** For `{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}`, the output should be `12, 15, and 18` for each column.

8. Find Saddle Point in Matrix

- A saddle point in a matrix is an element that is the smallest in its row but the largest in its column. Write a program to find all saddle points in a 2D matrix.
- **Example:** For `{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}`, there may or may not be a saddle point depending on the matrix.

9. Sort Each Row of a 2D Array

- Write a program to sort each row of a 2D array in ascending order.
 - **Example:** For `{{3, 2, 1}, {6, 5, 4}, {9, 8, 7}}`, the output should be `{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}`.
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3D Arrays

1. Sum of All Elements in a 3D Array

- Write a program to calculate the sum of all elements in a 3D integer array.
- **Example:** For a 3D array `{{{1, 2}, {3, 4}}, {{5, 6}, {7, 8}}}`, the output should be `36`.

2. Count Negative Numbers in 3D Array

- Write a function that counts the total number of negative elements in a 3D integer array.
- **Example:** For `{{{-1, 2}, {3, -4}}, {{5, -6}, {-7, 8}}}`, the output should be `4`.

3. Find Maximum Element in 3D Array

- Write a program that finds and returns the maximum element in a 3D integer array.
- **Example:** For `{{{1, 2}, {3, 4}}, {{5, 6}, {7, 8}}}`, the output should be `8`.

4. Check if All Elements in a Layer are Positive

- Write a function that checks if all elements in a specific layer of a 3D array are positive. The function should take the 3D array and layer index as inputs.
- **Example:** For array `{{{1, 2}, {3, 4}}, {{5, -6}, {7, 8}}}`, checking the second layer should return `false`.

5. Count Layers with Sum Greater than a Given Value

- Write a function that counts the number of layers in a 3D array where the sum of elements in each layer is greater than a specified value.
- **Example:** For array `{{{1, 2}, {3, 4}}, {{5, 6}, {7, 8}}}` and value `10`, the output should be `2`.

6. Flatten a 3D Array into 1D

- Write a program that converts a 3D array into a 1D array, keeping the elements in the same order.
- **Example:** For `{{{1, 2}, {3, 4}}, {{5, 6}, {7, 8}}}`, the output should be `[1, 2, 3, 4, 5, 6, 7, 8]`.

7. Find Average of Each Layer in 3D Array

- Write a program to find the average of elements in each layer of a 3D array and print it.
- **Example:** For $\{\{\{1, 2\}, \{3, 4\}\}, \{\{5, 6\}, \{7, 8\}\}\}$, the output should be 2.5 for the first layer and 6.5 for the second layer.

8. Find Minimum Element in Each Layer

- Write a function to find and return the minimum element in each layer of a 3D array.
- **Example:** For $\{\{\{3, 4\}, \{2, 1\}\}, \{\{6, 7\}, \{8, 5\}\}\}$, the output should be 1 for the first layer and 5 for the second layer.

9. Sum of Diagonal Elements in 3D Cubic Array

- Write a program to calculate the sum of diagonal elements of each 2D matrix in a 3D cubic array (only if the layers are square matrices).
- **Example:** For $\{\{\{1, 2\}, \{3, 4\}\}, \{\{5, 6\}, \{7, 8\}\}\}$, the output should be 5 for the first layer and 13 for the second.

10. Find Specific Value in 3D Array

- Write a function that checks if a specific integer exists in a 3D array and returns its coordinates (layer, row, and column).
- **Example:** For $\{\{\{1, 2\}, \{3, 4\}\}, \{\{5, 6\}, \{7, 8\}\}\}$ and value 6, the output should be (1, 1, 1) (layer 1, row 1, column 1).